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Batch: T22
Roll No.: -2401096
Subject:ASD&D

Experiment No. 02

Aim: To containerise an application using Docker.

LO 04: To automate provisioning, configuration, container orchestration, and cloud resource management using Ansible, Docker, Kubernetes, and Terraform.

Theory:

Docker

Docker is a platform/tool that lets you package an application along with everything it needs (code, libraries, dependencies, settings) into a single unit that can run the same way on any machine — your laptop, a server, the cloud, doesn't matter.

Containerization

This is the technique itself -packaging an application and its dependencies together into an isolated unit called a "container," so it runs consistently everywhere without "it works on my machine" problems. It's an alternative to running apps directly on a bare OS or inside a full virtual machine. Containers are lighter than VMs because they share the host machine's OS kernel instead of running a whole separate OS.

Image vs Container

Image	Container
A static, read-only template/blueprint	A running (or stopped) instance made from that image
Doesn't change, doesn't run by itself	Live, active, has a running process, can change while running
Similar to a class in programming	Similar to an object/instance of that class
Example: ubuntu image sitting in docker images	Example: container created using docker run ubuntu /bin/bash
Stored on disk as reusable layers	Created from an image at runtime with its own writable layer

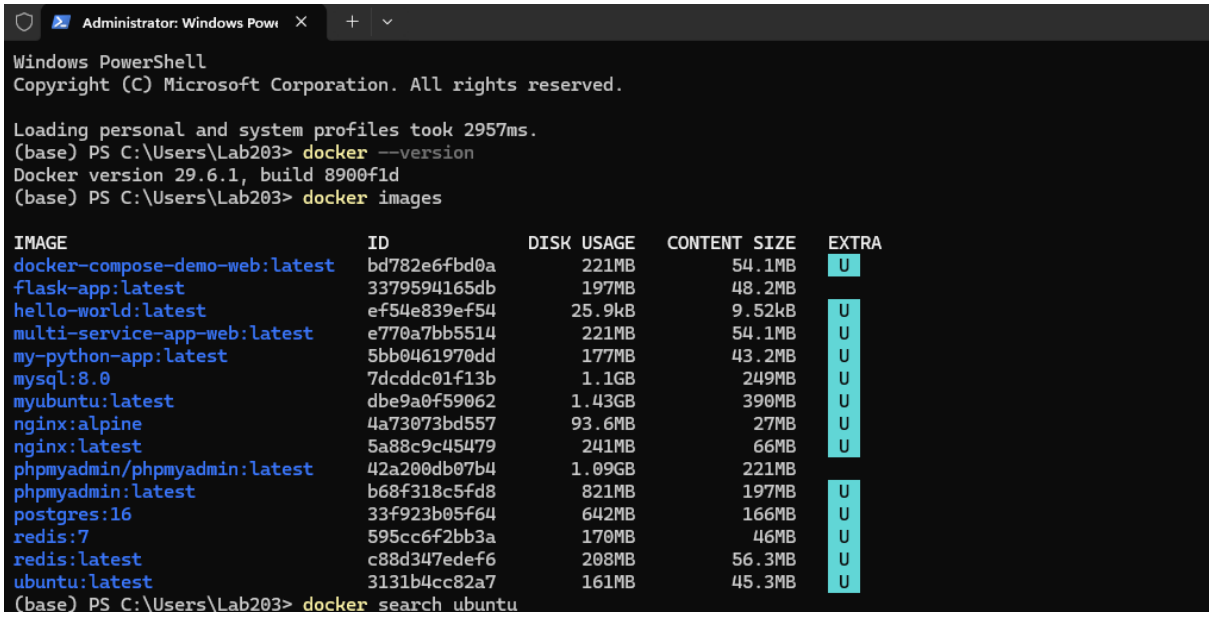
Commands Performed:

Sr. No.	Command	Purpose
1	docker --version	Displays the installed Docker Engine version, confirming Docker is installed and accessible from the command line.
2	docker images	Lists all Docker images currently stored locally.
3	docker search ubuntu	Searches Docker Hub for Ubuntu images.
4	docker search ubuntu --filter stars=50	Searches Ubuntu images having 50 or more stars.
5	docker pull ubuntu	Downloads the latest official Ubuntu image from Docker Hub.
6	docker images	Re-run after the pull to verify the Ubuntu image is available locally.
7	docker run ubuntu /bin/bash	Creates and starts a new container from the Ubuntu image and runs the Bash shell.
8	docker ps	Lists all currently running containers.

Explanation of Workflow

- The Docker version was first verified to confirm a working installation.
- Existing local images were listed using docker images.
- Docker Hub was searched for the Ubuntu repository.
- The search was refined using the --filter stars=50 option.
- The official Ubuntu image was pulled using docker pull ubuntu.
- docker images was executed again to verify the download.
- A container was created using docker run ubuntu /bin/bash.

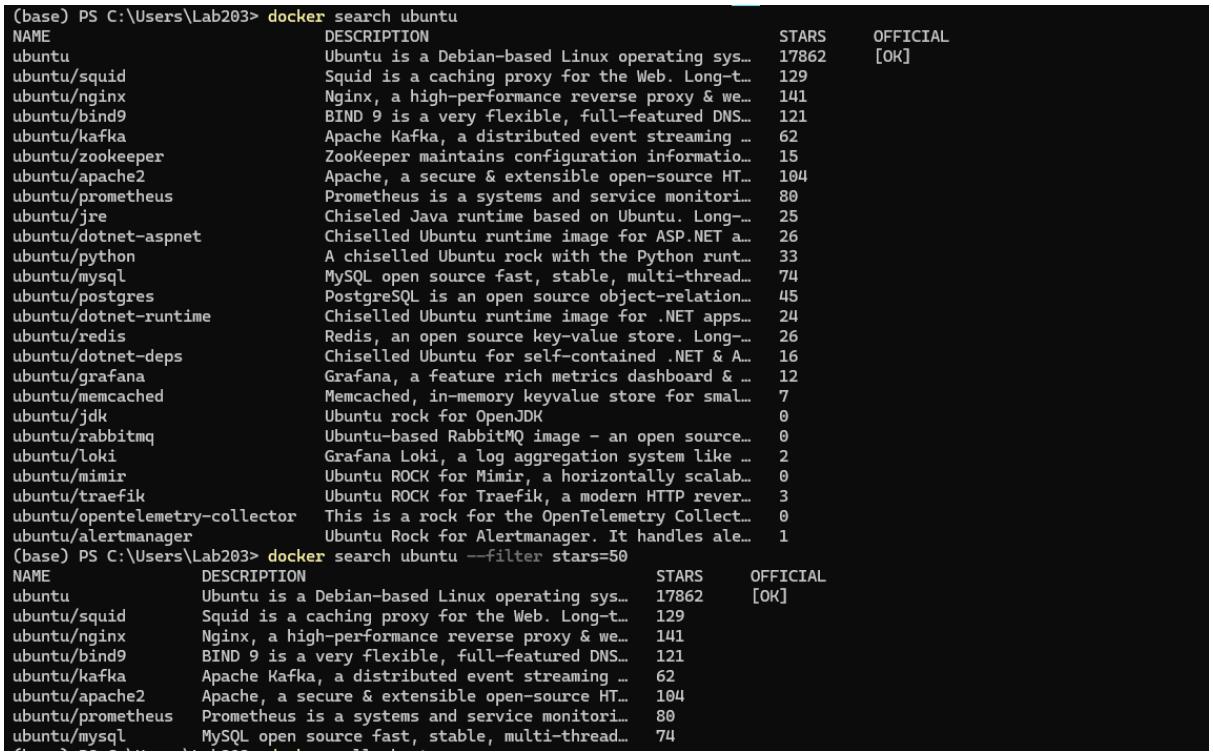
- docker ps confirmed that the container was running successfully.



```
Administrator: Windows Powe
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Loading personal and system profiles took 2957ms.
(base) PS C:\Users\Lab203> docker --version
Docker version 29.6.1, build 8900f1d
(base) PS C:\Users\Lab203> docker images

IMAGE                              ID             DISK USAGE   CONTENT SIZE   EXTRA
docker-compose-demo-web:latest     bd782e6fbd0a       221MB         54.1MB    U
flask-app:latest                   3379594165db       197MB         48.2MB
hello-world:latest                 ef54e839ef54      25.9kB         9.52kB    U
multi-service-app-web:latest       e770a7bb5514       221MB         54.1MB    U
my-python-app:latest               5bb0461970dd       177MB         43.2MB    U
mysql:8.0                          7dcddc01f13b       1.1GB          249MB    U
myubuntu:latest                    dbe9a0f59062      1.43GB          390MB    U
nginx:alpine                       4a73073bd557      93.6MB           27MB    U
nginx:latest                       5a88c9c45479       241MB           66MB    U
phpmyadmin/phpmyadmin:latest       42a200db07b4      1.09GB          221MB
phpmyadmin:latest                  b68f318c5fd8       821MB          197MB    U
postgres:16                        33f923b05f64       642MB          166MB    U
redis:7                            595cc6f2bb3a       170MB           46MB    U
redis:latest                       c88d347edef6       208MB         56.3MB    U
ubuntu:latest                      3131b4cc82a7       161MB         45.3MB    U
(base) PS C:\Users\Lab203> docker search ubuntu
```



```
(base) PS C:\Users\Lab203> docker search ubuntu
NAME                              DESCRIPTION                                     STARS     OFFICIAL
ubuntu                            Ubuntu is a Debian-based Linux operating sys…   17862     [OK]
ubuntu/squid                      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                      Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                      BIND 9 is a very flexible, full-featured DNS…   121
ubuntu/kafka                      Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper                  ZooKeeper maintains configuration informatio…   15
ubuntu/apache2                    Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus                 Prometheus is a systems and service monitori…   80
ubuntu/jre                        Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet              Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                     A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                      MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                   PostgreSQL is an open source object-relation…   45
ubuntu/dotnet-runtime             Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/redis                      Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-deps                Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                    Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                  Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                        Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                   Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/loki                       Grafana Loki, a log aggregation system like …   2
ubuntu/mimir                      Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/traefik                    Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector    This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager               Ubuntu Rock for Alertmanager. It handles ale…   1
(base) PS C:\Users\Lab203> docker search ubuntu --filter stars=50
NAME                DESCRIPTION                                     STARS     OFFICIAL
ubuntu              Ubuntu is a Debian-based Linux operating sys…   17862     [OK]
ubuntu/squid        Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx        Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9        BIND 9 is a very flexible, full-featured DNS…   121
ubuntu/kafka        Apache Kafka, a distributed event streaming …   62
ubuntu/apache2      Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus   Prometheus is a systems and service monitori…   80
ubuntu/mysql        MySQL open source fast, stable, multi-thread…   74
```

```
(base) PS C:\Users\Lab203> docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu
(base) PS C:\Users\Lab203> docker images

IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
docker-compose-demo-web:latest       bd782e6fbd0a       221MB         54.1MB    U
flask-app:latest                     3379594165db       197MB         48.2MB
hello-world:latest                   ef54e839ef54      25.9kB         9.52kB    U
multi-service-app-web:latest         e770a7bb5514       221MB         54.1MB    U
my-python-app:latest                 5bb0461970dd       177MB         43.2MB    U
mysql:8.0                            7dcddc01f13b       1.1GB          249MB    U
myubuntu:latest                      dbe9a0f59062      1.43GB          390MB    U
nginx:alpine                         4a73073bd557      93.6MB           27MB    U
nginx:latest                         5a88c9c45479       241MB           66MB    U
phpmyadmin/phpmyadmin:latest         42a200db07b4      1.09GB          221MB
phpmyadmin:latest                    b68f318c5fd8       821MB          197MB    U
postgres:16                          33f923b05f64       642MB          166MB    U
redis:7                              595cc6f2bb3a       170MB           46MB    U
redis:latest                         c88d347edef6       208MB         56.3MB    U
ubuntu:latest                        3131b4cc82a7       161MB         45.3MB    U
```

```
(base) PS C:\Users\Lab203> docker run ubuntu/bin/bash
Unable to find image 'ubuntu/bin/bash:latest' locally

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: pull access denied for ubuntu/bin/bash, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
(base) PS C:\Users\Lab203> docker run ubuntu /bin/bash
(base) PS C:\Users\Lab203> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
(base) PS C:\Users\Lab203>
```

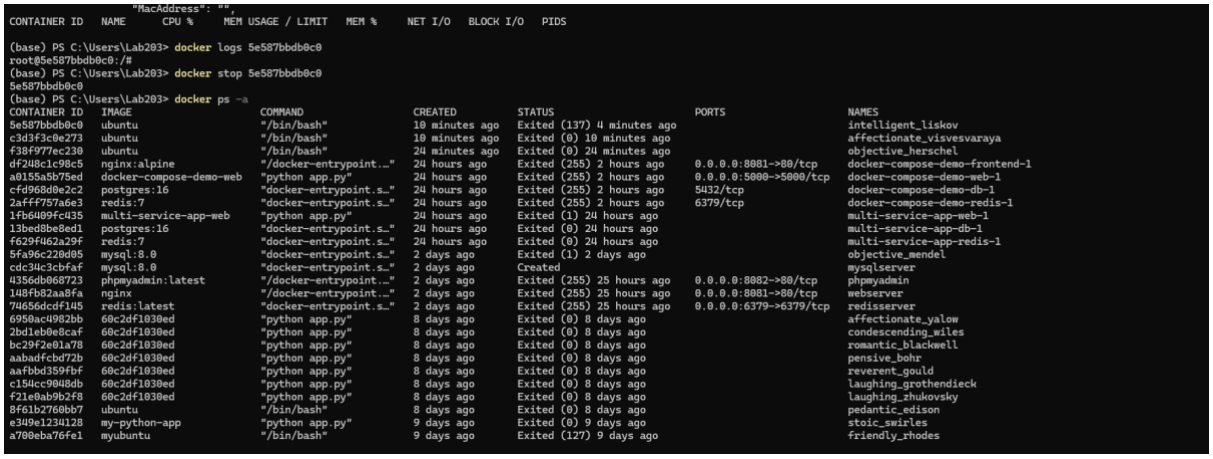
Additional Commands:

Command	Purpose
docker exec -it <container_id> bash	Open an interactive Bash terminal inside a running container.
docker inspect <container_id>	Display detailed information about a container in JSON format.
docker stats	Display live CPU, memory, network, and disk usage of running containers.
docker logs <container_id>	Display the logs generated by a container.
docker ps -a	List all containers, including running and stopped containers.
docker stop <container_id>	Stop a running container.
docker start <container_id>	Start a stopped container.

Command	Purpose
docker rm -f <container_id>	Forcefully remove a container.
docker rmi <image_id>	Remove a Docker image.
docker network ls	List all Docker networks.
docker volume ls	List all Docker volumes.
docker system prune	Remove unused containers, images, networks, and build cache to free disk space.

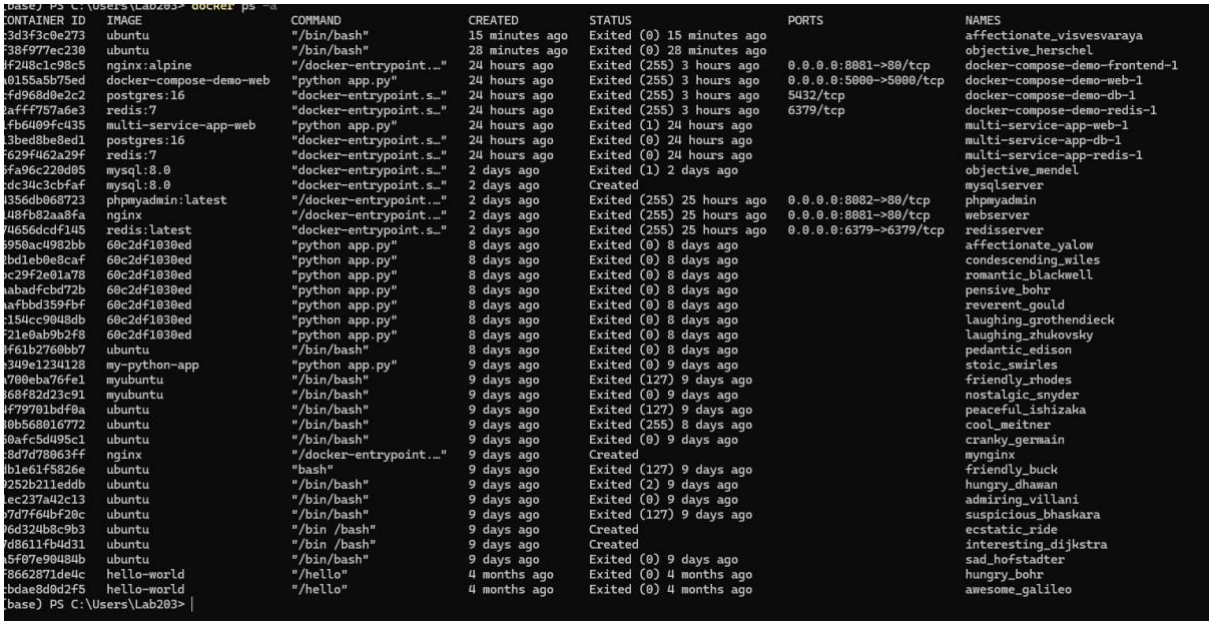


```
oading personal and system profiles took 2164ms.
(base) PS C:\Users\Lab203> docker inspect 5e587bbdb0c0
[
    {
        "Id": "5e587bbdb0c0139e6468bef1a11891f97c4f0f580830f5801bac12ab0563429f",
        "Created": "2026-07-30T02:47:47.628921583Z",
        "Path": "/bin/bash",
        "Args": [],
        "State": {
            "Status": "exited",
            "Running": false,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 0,
            "ExitCode": 137,
            "Error": "",
            "StartedAt": "2026-07-30T02:47:47.6806521152",
            "FinishedAt": "2026-07-30T02:53:11.5147584182"
        },
        "Image": "sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb",
        "ResolvConfPath": "/var/lib/docker/containers/5e587bbdb0c0139e6468bef1a11891f97c4f0f580830f5801bac12ab0563429f/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/5e587bbdb0c0139e6468bef1a11891f97c4f0f580830f5801bac12ab0563429f/hostname",
        "HostsPath": "/var/lib/docker/containers/5e587bbdb0c0139e6468bef1a11891f97c4f0f580830f5801bac12ab0563429f/hosts",
        "LogPath": "/var/lib/docker/containers/5e587bbdb0c0139e6468bef1a11891f97c4f0f580830f5801bac12ab0563429f/5e587bbdb0c0139e6468bef1a11891f97c4f0f580830f5801bac12ab0563429f-json.log",
        "Name": "/intelligent_liskov",
        "RestartCount": 0,
        "Driver": "overlayfs",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "bridge",
            "PortBindings": {},
            "RestartPolicy": {
                "Name": "no",
                "MaximumRetryCount": 0
            },
            "AutoRemove": false,
```

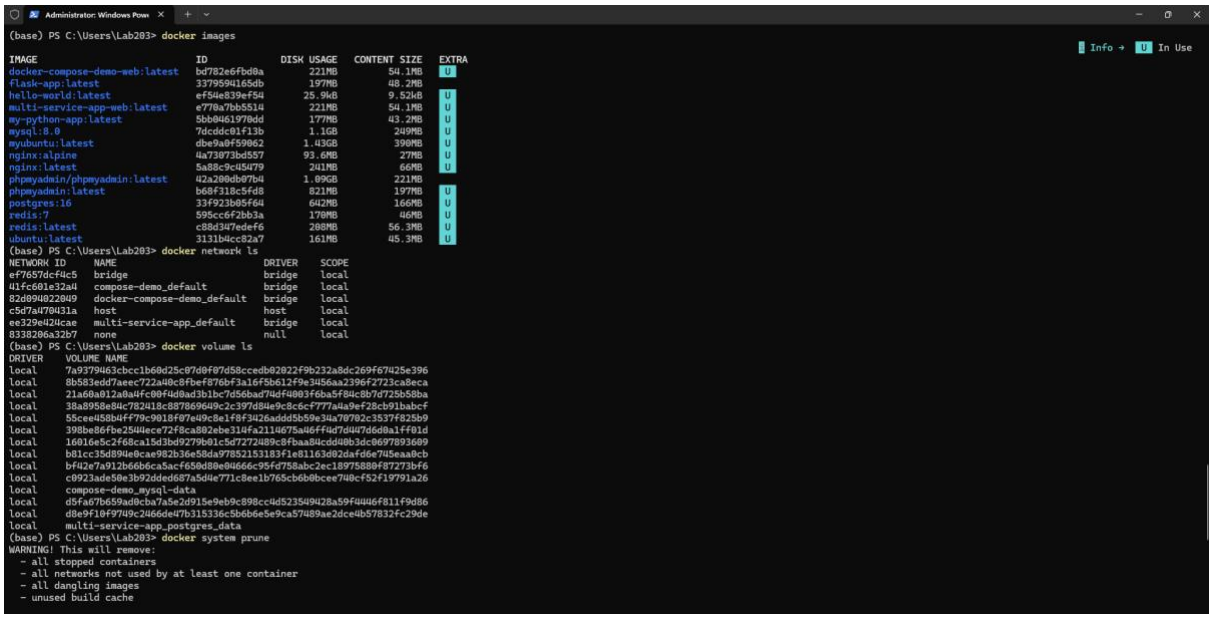


```
            "MacAddress": "",
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS

(base) PS C:\Users\Lab203> docker logs 5e587bbdb0c0
root@5e587bbdb0c0:/#
(base) PS C:\Users\Lab203> docker stop 5e587bbdb0c0
5e587bbdb0c0
(base) PS C:\Users\Lab203> docker ps -a
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS                       PORTS                    NAMES
5e587bbdb0c0   ubuntu                 "/bin/bash"              10 minutes ago  Exited (137) 4 minutes ago                            intelligent_liskov
c3d3f3c0e273   ubuntu                 "/bin/bash"              10 minutes ago  Exited (0) 10 minutes ago                             affectionate_visvesvaraya
f38f977ec230   ubuntu                 "/bin/bash"              24 minutes ago  Exited (0) 24 minutes ago                             objective_herschel
df248c1c98c5   nginx:alpine           "/docker-entrypoint.…"   24 hours ago    Exited (255) 2 hours ago     0.0.0.0:8081->80/tcp     docker-compose-demo-frontend-1
a0155a5b75ed   docker-compose-demo-web  "python app.py"        24 hours ago    Exited (255) 2 hours ago     0.0.0.0:5000->5000/tcp   docker-compose-demo-web-1
cfd968d0e2c2   postgres:16            "docker-entrypoint.s…"   24 hours ago    Exited (255) 2 hours ago     5432/tcp                 docker-compose-demo-db-1
2afff757a6e3   redis:7                "docker-entrypoint.s…"   24 hours ago    Exited (255) 2 hours ago     6379/tcp                 docker-compose-demo-redis-1
1fb6409fc435   multi-service-app-web  "python app.py"          24 hours ago    Exited (1) 24 hours ago                               multi-service-app-web-1
13bed8be8ed1   postgres:16            "docker-entrypoint.s…"   24 hours ago    Exited (0) 24 hours ago                               multi-service-app-db-1
f629f162a29f   redis:7                "docker-entrypoint.s…"   24 hours ago    Exited (0) 24 hours ago                               multi-service-app-redis-1
5fa96c220d05   mysql:8.0              "docker-entrypoint.s…"   2 days ago      Exited (1) 2 days ago                                 objective_mendel
cdc34c3cbfaf   mysql:8.0              "docker-entrypoint.s…"   2 days ago      Created                                               mysqlserver
4356db068723   phpmyadmin:latest      "/docker-entrypoint.…"   2 days ago      Exited (255) 25 hours ago    0.0.0.0:8082->80/tcp     phpmyadmin
148fb82aa8fa   nginx                  "/docker-entrypoint.…"   2 days ago      Exited (255) 25 hours ago    0.0.0.0:8081->80/tcp     webserver
74656dcdf145   redis:latest           "docker-entrypoint.s…"   2 days ago      Exited (255) 25 hours ago    0.0.0.0:6379->6379/tcp   redisserver
6950ac4982bb   60c2df1030ed           "python app.py"          8 days ago      Exited (0) 8 days ago                                 affectionate_yalow
2bd1eb0e8caf   60c2df1030ed           "python app.py"          8 days ago      Exited (0) 8 days ago                                 condescending_wiles
bc29f2e01a78   60c2df1030ed           "python app.py"          8 days ago      Exited (0) 8 days ago                                 romantic_blackwell
aabadfcbd72b   60c2df1030ed           "python app.py"          8 days ago      Exited (0) 8 days ago                                 pensive_bohr
aafbbd359fbf   60c2df1030ed           "python app.py"          8 days ago      Exited (0) 8 days ago                                 reverent_gould
c154cc9048db   60c2df1030ed           "python app.py"          8 days ago      Exited (0) 8 days ago                                 laughing_grothendieck
f21e0ab9b2f8   60c2df1030ed           "python app.py"          8 days ago      Exited (0) 8 days ago                                 laughing_zhukovsky
8f61b2760bb7   ubuntu                 "/bin/bash"              8 days ago      Exited (0) 8 days ago                                 pedantic_edison
e349e1234128   my-python-app          "python app.py"          9 days ago      Exited (0) 9 days ago                                 stoic_swirles
a700eba76fe1   myubuntu               "/bin/bash"              9 days ago      Exited (127) 9 days ago                               friendly_rhodes
```



```
(base) PS C:\Users\Lab203> docker ps -a
CONTAINER ID   IMAGE                      COMMAND                  CREATED          STATUS                      PORTS                    NAMES
e3d3f3c0e273   ubuntu                     "/bin/bash"              15 minutes ago   Exited (0) 15 minutes ago                            affectionate_visvesvaraya
f38f977ec230   ubuntu                     "/bin/bash"              28 minutes ago   Exited (0) 28 minutes ago                            objective_herschel
4f248c1c98c5   nginx:alpine               "/docker-entrypoint.…"   24 hours ago     Exited (255) 3 hours ago    0.0.0.0:8081->80/tcp     docker-compose-demo-frontend-1
a0155a5b75ed   docker-compose-demo-web    "python app.py"          24 hours ago     Exited (255) 3 hours ago    0.0.0.0:5000->5000/tcp   docker-compose-demo-web-1
5fd968d0e2c2   postgres:16                "docker-entrypoint.s…"   24 hours ago     Exited (255) 3 hours ago    5432/tcp                 docker-compose-demo-db-1
2afff757a6e3   redis:7                    "docker-entrypoint.s…"   24 hours ago     Exited (255) 3 hours ago    6379/tcp                 docker-compose-demo-redis-1
1fb6409fc435   multi-service-app-web      "python app.py"          24 hours ago     Exited (1) 24 hours ago                              multi-service-app-web-1
13bed8be8ed1   postgres:16                "docker-entrypoint.s…"   24 hours ago     Exited (0) 24 hours ago                              multi-service-app-db-1
f629f462a29f   redis:7                    "docker-entrypoint.s…"   24 hours ago     Exited (0) 24 hours ago                              multi-service-app-redis-1
6fa96c220d05   mysql:8.0                  "docker-entrypoint.s…"   2 days ago       Exited (1) 2 days ago                                objective_mendel
5dc34c3cbfaf   mysql:8.0                  "docker-entrypoint.s…"   2 days ago       Created                                              mysqlserver
14356db068723   phpmyadmin:latest          "/docker-entrypoint.…"   2 days ago       Exited (255) 25 hours ago   0.0.0.0:8082->80/tcp     phpmyadmin
1148fb82aa8fa   nginx                      "/docker-entrypoint.…"   2 days ago       Exited (255) 25 hours ago   0.0.0.0:8081->80/tcp     webserver
974656dcdf145   redis:latest               "docker-entrypoint.s…"   2 days ago       Exited (255) 25 hours ago   0.0.0.0:6379->6379/tcp   redisserver
56950ac4982bb   60c2df1030ed               "python app.py"          8 days ago       Exited (0) 8 days ago                                affectionate_yalow
72bd1eb0e8caf   60c2df1030ed               "python app.py"          8 days ago       Exited (0) 8 days ago                                condescending_wiles
6c29f2e01a78   60c2df1030ed               "python app.py"          8 days ago       Exited (0) 8 days ago                                romantic_blackwell
1abadfcbd72b   60c2df1030ed               "python app.py"          8 days ago       Exited (0) 8 days ago                                pensive_bohr
1afbbd359fbf   60c2df1030ed               "python app.py"          8 days ago       Exited (0) 8 days ago                                reverent_gould
e154cc9048db   60c2df1030ed               "python app.py"          8 days ago       Exited (0) 8 days ago                                laughing_grothendieck
f21e0ab9b2f8   60c2df1030ed               "python app.py"          8 days ago       Exited (0) 8 days ago                                laughing_zhukovsky
0f61b2760bb7   ubuntu                     "/bin/bash"              8 days ago       Exited (0) 8 days ago                                pedantic_edison
e3349e1234128   my-python-app              "python app.py"          9 days ago       Exited (0) 9 days ago                                stoic_swirles
1700eba76fe1   myubuntu                   "/bin/bash"              9 days ago       Exited (127) 9 days ago                              friendly_rhodes
3668f82d23c91   myubuntu                   "/bin/bash"              9 days ago       Exited (0) 9 days ago                                nostalgic_snyder
14f79701bdf0a   ubuntu                     "/bin/bash"              9 days ago       Exited (127) 9 days ago                              peaceful_ishizaka
80b5680167772   ubuntu                     "/bin/bash"              9 days ago       Exited (255) 8 days ago                              cool_meitner
160afc5d4951c1   ubuntu                     "/bin/bash"              9 days ago       Exited (0) 9 days ago                                cranky_germain
e8d7d78063ff   nginx                      "/docker-entrypoint.…"   9 days ago       Created                                              mynginx
4db1e61f5826e   ubuntu                     "bash"                   9 days ago       Exited (127) 9 days ago                              friendly_buck
17252b211eddb   ubuntu                     "/bin/bash"              9 days ago       Exited (2) 9 days ago                                hungry_dhawan
1ec237a42c13   ubuntu                     "/bin/bash"              9 days ago       Exited (0) 9 days ago                                admiring_villani
97d7f64bf20c   ubuntu                     "/bin/bash"              9 days ago       Exited (127) 9 days ago                              suspicious_bhaskara
196d324b8c9b3   ubuntu                     "/bin /bash"             9 days ago       Created                                              ecstatic_ride
7d8611fb4d31   ubuntu                     "/bin /bash"             9 days ago       Created                                              interesting_dijkstra
45f07e90484b   ubuntu                     "/bin/bash"              9 days ago       Exited (0) 9 days ago                                sad_hofstadter
f8662871de4c   hello-world                "/hello"                 4 months ago     Exited (0) 4 months ago                              hungry_bohr
cbdae8d0d2f5   hello-world                "/hello"                 4 months ago     Exited (0) 4 months ago                              awesome_galileo
(base) PS C:\Users\Lab203>
```



```
Administrator: Windows Pow
(base) PS C:\Users\Lab203> docker images
                                                                                                                    Info →  U  In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
docker-compose-demo-web:latest       bd782e6fbd8a       221MB         54.1MB    U
flask-app:latest                     3379594165db       197MB         48.2MB
hello-world:latest                   ef54e839ef54       25.9kB        9.52kB    U
multi-service-app-web:latest         e770a7bb5514       221MB         54.1MB    U
my-python-app:latest                 5bb0461970dd       177MB         43.2MB    U
mysql:8.0                            7dcddc01f13b       1.1GB          249MB    U
myubuntu:latest                      dbe9a0f59062       1.43GB         390MB    U
nginx:alpine                         4a73073bd557       93.6MB          27MB    U
nginx:latest                         5a88c9c45479       241MB           66MB    U
phpmyadmin/phpmyadmin:latest         42a200db07b4       1.09GB         221MB
phpmyadmin:latest                    b68f318c5fd8       821MB          197MB    U
postgres:16                          33f923b05f64       642MB          166MB    U
redis:7                              595cc6f2bb3a       170MB           46MB    U
redis:latest                         c80d3f7edef6       280MB          56.3MB    U
ubuntu:latest                        3131b4cc82a7       161MB          45.3MB    U
(base) PS C:\Users\Lab203> docker network ls
NETWORK ID     NAME                          DRIVER    SCOPE
ef7657dcf4c5   bridge                        bridge    local
41fc601e32a4   compose-demo_default          bridge    local
82d094022049   docker-compose-demo_default   bridge    local
c5d7a479431a   host                          host      local
ee329e424cae   multi-service-app_default     bridge    local
8338206a32b7   none                          null      local
(base) PS C:\Users\Lab203> docker volume ls
DRIVER    VOLUME NAME
local     7a9379463cbcc1b60d25c87d0f87d58ccedb02022f9b232a8dc269f67425e396
local     8b583edd7aeec722a40c8fbef876bf3a16f5b612f9e3456aa2396f2723ca8eca
local     21a60a012a0a4fc00f4d0ad3b1bc7d56bad74df4803f6ba5f84c8b7d725b58ba
local     38a8958e84c782418c887869649c2c397d84e9c8c6cf777a4a9ef28cb91babcf
local     55cee458b4ff79c9018f07e49c8e1f8f3426addd5b59e34a70702c3537f825b9
local     398be86fbe2544ece72f8ca802ebe3114fa21146754a46ff4d7d447d6d0a1ff01d
local     16016e5c2f68ca15d3bd9279b01c5d72724489c8fbaa84cdd40b3dc0697893609
local     b81cc35d894e0cae982b36e58da978521531831e81163d02dafd6e745eaa8cb
local     bf42e7a912b66b6ca5acf650d80e04666c954f758abc2ec18975880f872773bf6
local     c0923ade50e3b92dded687a5d4e771c8ee1b765cb6b0bcee740cf52f19791a26
local     compose-demo_mysql-data
local     d5fa67b659ad0cba7a5e2d915e9eb9c898cc4d523549428a59f44046f811f9d86
local     d8e9f10f9749c2466de47b315336c5b6b6e5e9ca57489ae2dce4b57832fc29de
local     multi-service-app_postgres_data
(base) PS C:\Users\Lab203> docker system prune
WARNING! This will remove:
  - all stopped containers
  - all networks not used by at least one container
  - all dangling images
  - unused build cache
```

Result

The Docker environment was successfully configured. The Ubuntu image was downloaded from Docker Hub, and a container was created and executed successfully using Docker commands. The running container was verified, and additional Docker commands were performed to manage containers, images, networks, and volumes.

Conclusion

In this experiment, Docker containerization was successfully implemented. The process of downloading a Docker image, creating a container, running an application environment, and managing Docker resources was studied. Docker provides a lightweight, portable, and consistent environment for deploying applications across different systems.